

L06: Relative Orientation; Fundamental & Essential Matrix

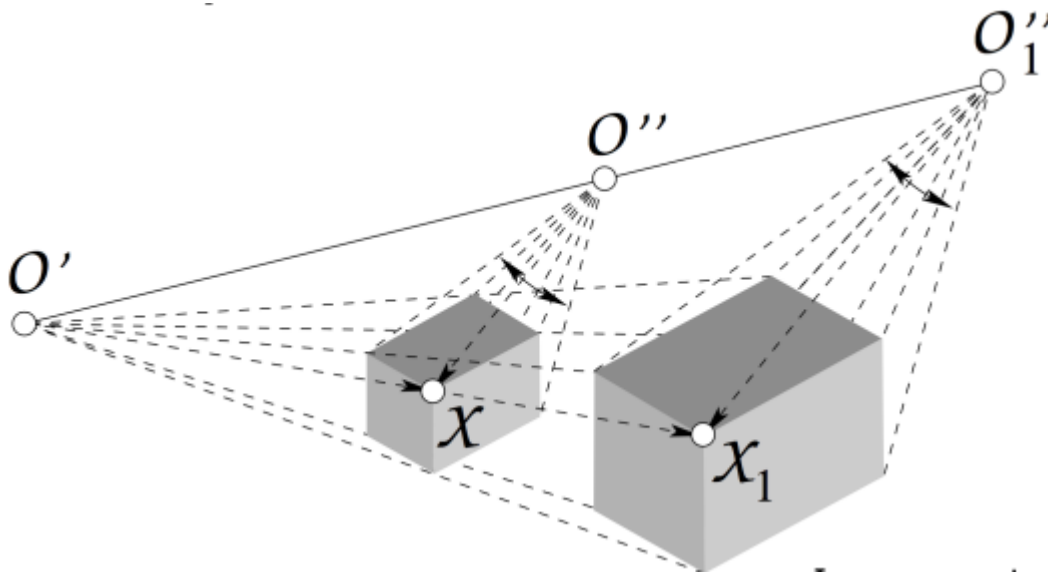
Camera Pair

- Two configurations - each - at which an image has been taken
 - A stereo camera (2 images at **same time**)
 - 1 Camera (takes 2 images after **moving**)

Orientation

- Orientation of camera pair:
 - Describable using independent orientations for each camera
- Calibrated cameras: **2 x 6 = 12** parameters
 - angle preserving mapping/model
 - Without additional information only obtainable:
 - $12 - 7 = 5$ **Parameters**
- Uncalibrated camera: **22** parameters
 - straight-line preserving mapping
 - Objects reconstructable up to a straight-line preserving mapping (projective transform, 15 DoF)
 - Given 2 images: only **obtain $22 - 15 = 7$ Parameters**
 - ≥ 5 points in 3D (15 DoF) for absolute orientation

What can be computed



- **Not** computable
 - Global translation
 - Global rotation
 - If cameras maintain their relative transformation
 - Scale
- Computable
 - **Rotation R** of the second camera w.r.t. the first one
 - 3 Parameters
 - **Direction B** of the line connecting to the centers of projection
 - 2 Parameters
 - Distance is unknown!

Photogrammetric Model

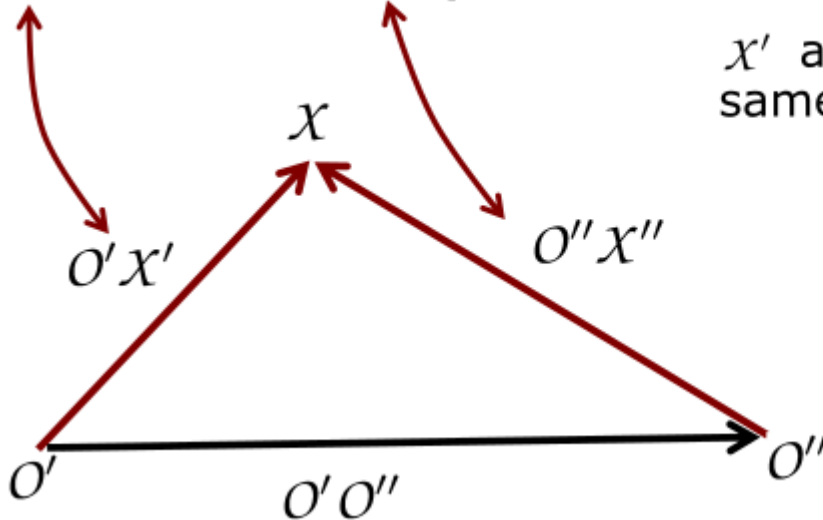
- Given images: object reconstructable only up to a **similarity transform**
- Orientation of Photogrammetric model called
 - **Absolute orientation**
 - At least **3 points in 3D**
 - (to estimate 7 parameters)

Cameras	# params per image	#params per img pair	# params for Relative Orientation	#params for absolute Orientation	minimum number of control points
Calibrated	6	12	5	7	3
Uncalibrated	11	22	7	15	5

Coplanarity Constraint for Straight-Line Preserving

- To obtain Fundamental Matrix
- intersection of 2 corresponding rays
 - express by triple spat product
 - $[A, B, C] = (A \times B) \cdot C$

$$[O'X' \quad O'O'' \quad O''X''] = 0$$



X' and X'' refer to the same point X in space

- Directions of vectors $O'X'$ and $O''X''$
 - ${}^n\mathbf{x}' = (\mathbf{R}')^{-1}(\mathbf{K}')^{-1}\mathbf{x}'$
 - where $\mathbf{x}'$: pixel coordinate in the image
 - as the normalised projection
 - ${}^n\mathbf{x}' = [\mathbf{I}_3 | -\mathbf{X}_{O'}]\mathbf{X}$
 - $\mathbf{X}$ in world coordinates
 - provides the direction from center of projection to the point in 3D
 - derivable from image coordinates $\mathbf{x}', \mathbf{x}''$
 - $\mathbf{x}' = \mathbf{P}'\mathbf{X}$
 - $\mathbf{x}'' = \mathbf{P}''\mathbf{X}$
 - with projection matrices
 - $\mathbf{P}' = \mathbf{K}'\mathbf{R}'[\mathbf{I}_3 | -\mathbf{X}_{O'}]$
 - $\mathbf{P}'' = \mathbf{K}''\mathbf{R}''[\mathbf{I}_3 | -\mathbf{X}_{O''}]$
- Base Vector $O'O''$:
 - $\mathbf{B} = \mathbf{X}_{O''} - \mathbf{X}_{O'}$
 - coordinates of projection centers

Using coplanarity constraint

- $[O'X' \quad O'O'' \quad O''X''] = 0$

- rewritten as

$$\begin{aligned} [{}^n\mathbf{x}' \quad \mathbf{b} \quad {}^n\mathbf{x}''] &= 0 \\ {}^n\mathbf{x}'(\mathbf{b} \times {}^n\mathbf{x}'') &= 0 \\ {}^n\mathbf{x}'^T \mathbf{S}_b {}^n\mathbf{x}'' &= 0 \end{aligned}$$

- S_b : skew-symmetric matrix:

- Cross product can be rewritten as a product of

- a skew-symmetric matrix
- a vector

- Correctness results from cross product as

$$\underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}_{\mathbf{b}} \times \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{x}} = \begin{bmatrix} -b_3x_2 + b_2x_3 \\ b_3x_1 - b_1x_3 \\ -b_2x_1 + b_1x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -b_3 & b_2 \\ b_3 & 0 & -b_1 \\ -b_2 & b_1 & 0 \end{bmatrix}}_{\mathbf{S}_b} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{x}}$$

$$\mathbf{x}'^T \underbrace{(\mathbf{K}')^{-T}(\mathbf{R}')^{-T} \mathbf{S}_b (\mathbf{R}'')^{-1} (\mathbf{K}'')^{-1}}_{\mathbf{F}} \mathbf{x}'' = 0$$

$$\mathbf{x}'^T \mathbf{F} \mathbf{x}'' = 0$$

- with F as the **Fundamental matrix**
- F contains all available information about the relative orientation of 2 uncalibrated images

Fundamental Matrix from the Camera Projection Matrices

- We have

$$\circ P' = [A' | \mathbf{a}'] = \left[\frac{K'R'}{A'} \mid \frac{-K'R'\mathbf{X}_{O'}}{\mathbf{a}'} \right]$$

- Recover Projection center

$$\circ \mathbf{X}_{O'} = \mathbf{A}'^{-1} \mathbf{a}'$$

- Base line:

$$\circ \mathbf{b}'_{12} = \mathbf{A}''^{-1} \mathbf{a}'' - \mathbf{A}'^{-1} \mathbf{a}'$$

- We can compute F :

$$\circ F = A'^{-T} S_{b'_{12}} A''^{-1}$$

$$\circ S_b = \begin{bmatrix} 0 & -b_3 & b_2 \\ b_3 & 0 & -b_1 \\ -b_2 & b_1 & 0 \end{bmatrix}$$

- Alternative definition

- $F_{21} = F_{12}^T$

Essential Matrix (For Calibrated Cameras)

- Calibrated cameras simplify the orientation problem
- Assumptions: no lens distortions or other imaging errors
- Exploiting the fundamental matrix

$$\frac{\mathbf{x}'^T (\mathbf{K}')^{-T} (\mathbf{R}')^{-T} \mathbf{S}_b (\mathbf{R}'')^{-1} (\mathbf{K}'')^{-1} \mathbf{x}''}{\mathbf{F}} = 0$$

$$\frac{\mathbf{x}'^T (\mathbf{K}')^{-T} (\mathbf{R}')^{-T} \mathbf{S}_b (\mathbf{R}'')^T (\mathbf{K}'')^{-1} \mathbf{x}''}{k_{\mathbf{x}'}^T} = 0$$

$$k_{\mathbf{x}'}^T \frac{\mathbf{R}' \mathbf{S}_b \mathbf{R}''^T}{\mathbf{E}} k_{\mathbf{x}''} = 0$$

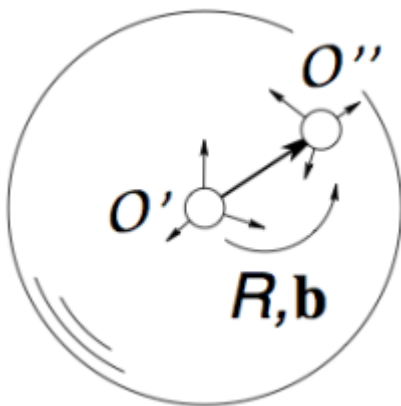
$${}^k \mathbf{x}'^T \mathbf{E} {}^k \mathbf{x}'' = 0$$

- Constraint for calibrated camera analogous to F
- F for calibrated camera: **Essential Matrix**
 - $\mathbf{E} = \mathbf{R}' \mathbf{S}_b \mathbf{R}''^T$
 - 5 DoF
 - relative orientation of the calibrated camera/image pair
- Matrix: 9 entries - 5 = **4 constraints**
- Essential matrix is **homogenous & singular**

Popular Parameterisations for Relative Orientation

- 5 parameters

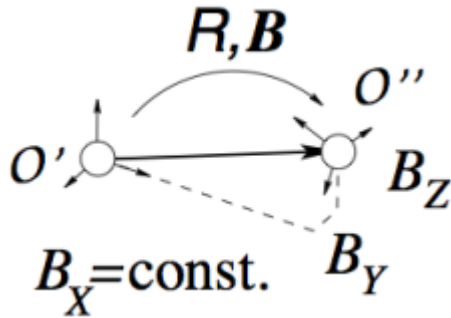
1. General Parameterisation of Dependent Images



-
- normalised direction vector $\mathbf{b}$
- rotation matrix $\mathbf{R}$
- ${}^k \mathbf{x}'^T \mathbf{S}_b \mathbf{R}^T {}^k \mathbf{x}'' = 0$

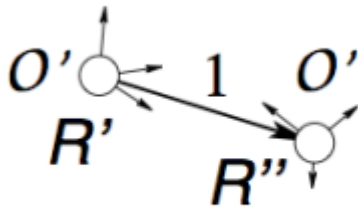
- with $|b| = 1$

2. Photogrammetric Parametrisation of Dependent Images



-
- Two components B_Y and B_Z of the base direction vector ($B_X = \text{const.}$)
- a rotation matrix R
- $\underbrace{(B_X, B_Y, B_Z)}_b, \underbrace{(\omega, \phi, \kappa)}_R$ with $B_X^2 + B_Y^2 + B_Z^2 = 1$

3. Parameterisation with Independent Images



-
- rotation matrix $R'(\omega', \phi', \kappa')$
- rotation matrix $R''(\omega'', \phi'', \kappa'')$
- fixed basis of constant length
- ${}^k x'^T R' S R''^T {}^k x'' = 0$
 - with $\omega', S = \text{const.}$